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Compressive learning scaffolds higher-order network structure to enhance human knowledge acquisition

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Abstract

Humans naturally seek knowledge, yet integrating vast, fragmented information remains challenging. Traditionally, knowledge acquisition has relied on random walks within network—an unguided and inefficient process. Here, we introduce “compressive learning”, a conceptual framework that embeds higher-order structural features—specifically node-degree inhomogeneity—into pre-learning trajectories to scaffold more efficient learning. We show that scale-free networks, owing to their pronounced degree inhomogeneity, are more compressible and more learnable than other network types. Critically, pre-learning paths that highlight this inhomogeneous higher-order structure facilitate subsequent network learning. Magnetoencephalography (MEG) recordings reveal that compressive pre-learning enhances structured neural representations in the dorsal anterior cingulate cortex (ACC). Two-stage computational modeling indicates that compressive learning constructs a network skeleton defined by higher-order structure that efficiently accommodates new inputs. Together, our results highlight the central role of higher-order network structure in human learning and offer a strategic approach to effectively integrating fragmented information into a coherent knowledge framework.

Introduction

Humans are inherently driven to seek knowledge. Yet in today's era of information explosion, we often become overwhelmed and lose sight of the bigger picture¹. Individuals face significant challenges in constructing integrated knowledge frameworks from data that far exceed their processing capacities^{2,3}. Understanding the cognitive and computational mechanisms that support knowledge acquisition—and developing efficient strategies to scaffold that process—is therefore critical for individuals and societies alike^{4–11}.

Knowledge learning can be conceptualized as navigation through a network, where discrete pieces of information and their relationships are represented as nodes and edges, respectively^{12–14}. From this perspective, learning involves uncovering the hidden structure of the network. This can be empirically studied using the network learning paradigm^{12,15–19}, where participants are exposed to sequences of arbitrary images embedded in a predefined network, and each edge dictates the transition probability between images (Figure 1a). As participants predict upcoming images through trial-and-error, improvements in accuracy reflect their acquisition of the underlying network structure.

While prior research has shown that specific abstract structure facilitates the learning process and supports generalization to new inputs^{9,12,16,20–23}, the core structural properties that determine a network's learnability remain unclear. Critically, conventional approaches often use random-walk paths for learning—bottom-up, unguided sampling that is inherently inefficient and slow. Here, we sought to develop an approach that embeds key structural properties into a pre-learning path to enhance later network learning, namely “compressive learning”. This idea also aligns with constructivist theory^{24,25}, which posits that learning is an active process where new information is gradually integrated into pre-existing mental models.

Recent work suggests that higher-order network structures—such as community organization and node-degree inhomogeneity—support the mental representation of knowledge networks^{14,17,26–28}. These properties are also widespread in real-world systems, further highlighting their role in efficient information storage^{29–33}. Here we focus on node-degree inhomogeneity, a key higher-order property of network structure. As illustrated in Figure 1b, a lattice network exhibits low inhomogeneity, with all nodes having the same number of connections. Random and small-world networks show moderate inhomogeneity, characterized by approximately normal node-degree distributions. In contrast, scale-free networks demonstrate the highest inhomogeneity,

following a long-tail distribution in which a few nodes possess a disproportionately large number of connections.

Inspired by the concept of “compressive sensing” in engineering^{34,35}, here we propose a “compressive learning” conceptual framework in the context of network learning. Specifically, we hypothesize that node-degree inhomogeneity, as a higher-order structural property, may help learners organize information around a relatively small subset of hub nodes, thereby facilitating the formation of an efficient representational scaffold for subsequent learning and inference. In this sense, “compression” refers to a potentially more efficient cognitive organization of relational structure that helps connect fragmented information during network learning, rather than a formally defined quantitative property of the network itself. Accordingly, the terms “compression” and “compressive” used throughout the manuscript should be understood as descriptive conceptual notions referring to scaffold-based organization of knowledge, rather than a formally established theory or quantification of network compressibility.

First, we hypothesize that higher-order structural properties like node-degree inhomogeneity are related to network compressibility—and in turn impact its learnability. One type of quantitative analysis in terms of four network types confirmed that scale-free networks are more compressible than others (Figure 1c), and that greater node-degree inhomogeneity reliably predicts higher compressibility across multiple quantification methods (Figure 1d). We thereby formulated Hypothesis I: scale-free network structure, due to its strong inhomogeneity, will exhibit superior learnability in humans compared to the other three network types, even when all networks embed the same image sets (Figure 1e).

Building on this, we proposed Hypothesis II, termed “compressive learning” (Figure 1f). We hypothesize that a pre-learning path designed to highlight the inhomogeneous structure of a network can establish a scaffold that facilitates subsequent learning. This approach departs from conventional random-walk network learning by embedding structure-based guidance early in the learning process, thereby acting as a compressive, structure-driven pre-training strategy, rather than a purely bottom-up, unguided process.

To test these ideas, we conducted two large-scale behavioral experiments ($N > 400$), which confirmed both the superior learnability of inhomogeneous (scale-free) networks and the effectiveness of the compressive learning approach. Magnetoencephalography (MEG) recordings further revealed that compressive pre-learning enhances the formation of structured neural

representations in the dorsal anterior cingulate cortex (ACC). Finally, we developed two-stage computational models that highlight the higher-order structure to account for compressive learning. Together, these findings underscore the important role of higher-order network structure in human learning and demonstrate how it can be strategically harnessed to effectively integrate fragmented information into coherent frameworks.

Results

Behavioral paradigm for knowledge network learning

The behavioral task was adapted from previous network learning paradigms¹⁷. A series of 16-node, undirected and unweighted networks were created in which the 1-step transition probabilities between individual nodes are pre-determined accordingly (Figure 2a). 16 black-and-white images were randomly assigned to the 16 nodes of the network (Figure 2a, left), constituting the to-be-learned artificial knowledge network. Next, image sequences were generated based on a random walk path within the network (Figure 2b, right). Subjects performed continuous prediction through trial-and-error without being told about the actual underlying network. On each trial, subjects were presented with a Cue image and predicted its succeeding image from the Target-Distractor image pair (Figure 2b, left), and the Target image on the current trial would automatically become the Cue image in the next trial (Figure 2b, middle). We monitored subjects' mouse trajectories during decision-making and performed a trial-wise binary classification analysis to quantify network learning performance (Figures 2c and S2). Specifically, the mouse cursor would be closer to the target than to the distractor if subjects had learnt the transitional relations. Moreover, we employed a hint system and a coin system to motivate learning (see details in Methods and Figure S1). Notably, different from implicit network learning paradigms^{15,29,36}, we employed an explicit learning paradigm in which participants predicted the next image based on learned pairwise transitions^{37,38}.

Inhomogeneity structure facilitates knowledge network learning

Experiment 1 ($N = 160$) directly tested Hypothesis I by examining whether a network with high node-degree inhomogeneity would demonstrate superior learnability, given its high

compressibility (see Figure 1e). To this end, we constructed four types of 16-node networks (Figure 2a): Lattice (pink), Random (green), Small-world (blue), Scale-free (orange). These networks are matched in the number of nodes (16), edges (32), and average node degree (4), but differ in node-degree inhomogeneity (Figures 2b and S1). Notably, the Lattice network exhibits the lowest inhomogeneity, with each node connected to exactly four others. The Random and Small-world networks display moderate inhomogeneity with approximately normal node-degree distributions. In contrast, the Scale-free network shows the highest inhomogeneity, characterized by a few highly connected nodes. The same set of 16 images was embedded into each network and shuffled across participants to avoid image-related confounds.

Subjects who passed tests within 5 attempts in the first session were invited to participate in the main experiment (see Methods for more details; the same applies to Experiments 2 and 3). A total of 160 participants who passed the prescreening were randomly assigned to one of the four network conditions (40 per group). Each group learned its assigned network through image sequences generated by random-walk paths within the respective network (Figure 2a). Participants completed 1,000 trials divided into five blocks. To assess learning performance, we conducted a trial-wise binary classification analysis of mouse trajectories (Figure 2c). As shown in Figure 2d, all groups exhibited above-chance prediction before hints, i.e., moving the mouse toward the target rather than the distractor. Crucially, the Scale-free group demonstrated the highest learning performance (Figures 2e and S3–S5). A generalized linear mixed-effects model (GLMM), which controlled for recency, node degree, and other potential confounders, confirmed the superiority of the Scale-free network (ANOVA from GLMM1 with likelihood ratio tests, main effect of network and trial numbers: $\chi^2(3) = 143.92, p < 0.001$, $\chi^2(1) = 26.71, p < 0.001$, interaction effect between network and trial numbers: $\chi^2(3) = 4.45, p = 0.216$; post-hoc comparisons with Bonferroni correction from GLMM1, two-sided, Scale-free vs. Lattice: $z = 10.63, p < 0.001$; Scale-free vs. Random: $z = 9.81, p < 0.001$; Scale-free vs. Small-world: $z = 8.68, p < 0.001$; see the full statistical reports in the Supplementary Information).

Notably, because the task was challenging, some participants who did not fully acquire the node relationships could respond correctly only after the hint appeared, resulting in lower pre-hint performance (Figure 2d). As the hint gradually reached its maximum luminance around 0.5 s post hint onset (or 1.3 s after target-distractor onset), some participants still showed relatively low performance even after 0.8 s, likely reflecting the delay between detecting the hint and executing

the response. Importantly, the key findings remained robust to this variability (see control analysis in Figure S6) and were not driven by a small subset of participants. Moreover, reaction time is not a reliable measure due to the hint system design. In addition to the mouse trajectory analysis, we also analyzed click accuracy from before-hint-response trials, which replicated the results of Experiment 1 (see Figure S8A). Notably, the four networks may also differ in other topological properties beyond inhomogeneity (e.g., clustering coefficient, characteristic path length; see Table S3 for comparisons), which could be systematically examined in future work.

In sum, these findings support Hypothesis I, suggesting that the compressibility of network structure—reflected in higher-order node-degree inhomogeneity—enhances network learnability in humans.

Compressive learning promotes knowledge network learning

Experiment 2 ($N = 240$) tested Hypothesis II (“compressive learning”) by examining whether emphasizing a network’s inhomogeneous structure during pre-learning could establish a core scaffold that facilitates subsequent random-walk learning (see Figure 1f). The Scale-free network was used, given its pronounced higher-order inhomogeneity, making it ideal for testing this hypothesis.

In a scale-free network, nodes can be ranked by node degree, from low-degree leaf nodes (small circles) to high-degree hub nodes (large circles) (Figure 3a, left). While typical random-walk paths (as used in Experiment 1) visit all nodes probabilistically, we designed pre-learning sequences that explicitly emphasized node-degree structure. Specifically, a 200-image sequence from a random walk (*RandomWalk*) was reordered based on node degree—either in descending (*HubToLeaf*) or ascending (*LeafToHub*) order—yielding three pre-learning paths: *RandomWalk*, *HubToLeaf*, and *LeafToHub* (Figures 3a and 3b). All three sequences during Learning I therefore contained the same images with identical frequencies, differing only in order.

Specifically, the *HubToLeaf* path introduced relationships between hubs and their neighbors first, followed by progressively lower-degree nodes. In contrast, *LeafToHub* began with leaf-node relationships, introducing hubs last. These reorderings disrupted the probabilistic continuity of the original random-walk path, instead highlighting the underlying inhomogeneity of the network to varying degrees. Based on the central role of hub-related substructures in network learning and the potential primacy effect in long-term memory learning^{39,40} (tens of minutes between Learning I

and Learning II), we hypothesized that the *HubToLeaf* path—by highlighting hub-related substructures during early phase of Learning I—would better scaffold the network structure, thereby facilitating subsequent learning.

A total of 238 new participants were randomly assigned to one of the three pre-learning conditions. Each group viewed an image sequence from one of the pre-learning paths (200 trials; Learning I), followed by the same *RandomWalk* sequence used in all groups (800 trials; Learning II) (Figure 3a, right). Critically, only Learning II performance was compared between groups to examine the pre-learning paths' impacts, and therefore any learning difference would arise from the influence of pre-learning path. As in Experiment 1, learning was assessed using trial-wise classification of mouse trajectories (see Figure S8B for similar results based on click accuracy from before-hint-response trials), and all groups performed above chance before receiving hints (Figure 3c). The 800 trials in Learning II were divided into eight bins (100 trials each) to analyze learning progression. As shown in Figure 3d, all groups began with similar performance, but the *HubToLeaf* group gradually surpassed the others and maintained this advantage (see GLMM2.1 in the Supplementary Information). Averaged performance across all Learning II trials followed a similar trend, confirming the superiority of the *HubToLeaf* condition.

Notably, the interaction between group and trial number indicates that the *HubToLeaf* group developed an advantage more rapidly than the others during the testing phase (Figure 3e; ANOVA from GLMM2.2 with likelihood ratio tests, main effect of group and trial numbers: $\chi^2(2) = 124.68$, $p < 0.001$, $\chi^2(1) = 22.71$, $p < 0.001$, interaction effect between group and trial numbers: $\chi^2(2) = 9.35$, $p = 0.009$; post-hoc comparisons with Bonferroni correction from GLMM2.2, two-sided, *HubToLeaf* vs. *RandomWalk*: $z = 10.73$, $p < 0.001$; *HubToLeaf* vs. *LeafToHub*: $z = 8.04$, $p < 0.001$; *RandomWalk* vs. *LeafToHub*: $z = -2.77$, $p = 0.017$). These findings were further replicated in another offline experiment (Figure S11). Moreover, as all the pre-learning sequences were generated from the same image sequences differing only in order, variation or repetition differences cannot account for the superior *HubToLeaf* performance. Consistently, frequency analyses of the 64 transition pairs related to hub and leaf nodes revealed no differences across the three groups (Figure S9).

We next examined whether the benefit of compressive learning is confined to specific pairs or extends to the entire network. To this end, we separated hub-related and leaf-related pairs in Learning II and compared accuracies across the three groups. As shown in Figure 3f, the

HubToLeaf group outperformed the *RandomWalk* group on both hub- and leaf-related pairs (Hub-related pairs: ANOVA for main effect of group: $\chi^2(2) = 170.34$, $p < 0.001$; post-hoc comparisons with Bonferroni correction, *HubToLeaf* vs. *RandomWalk*: $z = 9.912$, $p < 0.001$; *HubToLeaf* vs. *LeafToHub*: $z = 12.294$, $p < 0.001$; *RandomWalk* vs. *LeafToHub*: $z = 2.272$, $p = 0.070$; Leaf-related pairs: ANOVA for main effect of group: $\chi^2(2) = 45.90$, $p < 0.001$; post-hoc comparisons with Bonferroni correction, *HubToLeaf* vs. *RandomWalk*: $z = 5.104$, $p < 0.001$; *HubToLeaf* vs. *LeafToHub*: $z = -1.339$, $p = 0.542$; *RandomWalk* vs. *LeafToHub*: $z = -6.429$, $p < 0.001$), indicating that the enhancement induced by compressive learning is not restricted to hub nodes but generalizes across the network.

Interestingly, the *HubToLeaf* group showed an advantage over the *LeafToHub* group only for hub-related pairs (left violin plot in Figure 3f), but not for leaf-related ones (right violin plot in Figure 3f). We suggest that the *LeafToHub* group's initial emphasis on leaf pairs may have partially rescued their vulnerability, resulting in performance comparable to the *HubToLeaf* group. This interpretation aligns with previous findings showing that leaf-related edges are more susceptible to memory degradation and require greater compensation⁴¹. Crucially, however, the *LeafToHub* group never outperformed the *HubToLeaf* group, even on leaf-related pairs that were primed during the *LeafToHub* pre-learning path (see Figure S10 for analyses based on pairs initially learned in either path), confirming that the observed advantage is not simply a primacy effect tied to specific pairs, but reflects a broader network-level change. Moreover, in the *HubToLeaf* group, pairs experienced early during Learning I did not outperform those learned later (Figure S10), appearing inconsistent with a simple primacy account. This may reflect the dense learning demands associated with hub-related transitions (Figure S7), which could attenuate potential primacy advantages of early learned pairs.

In sum, the results support the “compressive learning” (Hypothesis II): briefly priming the inhomogeneity network structure (*HubToLeaf* pre-learning path) would facilitate subsequent random-walk network learning.

Dorsal ACC tracks network structure establishment facilitated by compressive learning

After confirming the effectiveness of “compressive learning”, we next examined the underlying neural mechanism using MEG recordings. A total of 40 new subjects were recruited and assigned randomly to the *HubToLeaf* (purple) or *LeafToHub* (green) groups (20 participants per group). The

experimental paradigm mirrored that of Experiment 2 with minor modifications to accommodate MEG constraints (Figures 4a and S1b). During Learning I, participants were exposed to either the *HubToLeaf* or *LeafToHub* image sequence, followed by an identical *RandomWalk* sequence across both groups for Learning II (600 trials). Crucially, only neural activity from Learning II was analyzed and compared between the groups. We focused on neural responses elicited by cue images, which would potentially reflect their positions in the to-be-learned network. Moreover, a modification from the design of Experiment 2 was the introduction of a blank period following the cue image offset to examine the neural dynamics of network representations. Building on our prior behavioral finding that the *HubToLeaf* path facilitates network formation relative to the *LeafToHub* path (Experiment 2), we sought to identify the corresponding neural signatures—that is, the time windows in which the *HubToLeaf* group exhibited stronger neural representations of the to-be-learned network than the *LeafToHub* group.

To probe neural representations of network structure, we conducted representational similarity analysis (RSA) on MEG responses following the presentation of cue images (Figure 4b). We hypothesized that the multivariate dissimilarities between cue-evoked neural patterns (neural representation dissimilarity matrices; neural RDMs) would reflect the transitional distances between nodes in the underlying network^{17,42} (model RDM, Figure 4c). Specifically, we included whole-brain sensor data as multivariate neural patterns to calculate neural RDMs to probe the representation dynamics^{17,43–46}. At each time point, the neural RDMs were regressed against the model RDMs, producing a time-resolved index of network representational strength. Due to random shuffling of image-node mappings across participants, these effects cannot be attributed to image identity. As shown in Figure 4d, both groups demonstrated above-chance neural representations of network structure (black line). Crucially, consistent with our hypothesis, during the blank interval preceding the target-distractor display, the *HubToLeaf* group exhibited significantly stronger network representations than the *LeafToHub* group (LMM1 with cluster-based permutation test, one-sided, $p = 0.042$). Notably, during the blank interval between cue offset and target-distractor onset, the *LeafToHub* group initially exhibited representational strength comparable to that of the *HubToLeaf* group, but this signal decayed more rapidly in later time bins (Figure S13). The early network representation observed during the blank period in both groups likely reflects rapid, stimulus-triggered mnemonic reactivation of the network structure. In contrast, a later component (~ 1.2 s), which was stronger in the *HubToLeaf* group, likely reflects the greater

persistence and robustness of structural representations formed through compressive learning.

Further analyses focused on this time window of interest (~1.2 s post-cue, shaded in orange) confirmed the robustness of this effect. First, multidimensional scaling (MDS) of neural RDMs revealed a hierarchical node organization resembling the ground-truth network (Figure 4e) for the *HubToLeaf* group (Figure 4f), but not for the *LeafToHub* group (Figure 4g). Second, the representational advantage in the *HubToLeaf* group was evident for both hub- and leaf-related substructures (Figure 4i; LMM1, one-sided, hub-related, $\beta = 0.012$, $t(44.48) = 2.12$, $p = 0.040$; leaf-related, $\beta = 0.007$, $t(43.85) = 2.25$, $p = 0.029$), indicating a network-wide rather than hub-specific effect. Moreover, both groups showed comparable neural encoding of node-degree information at the early stage but not at the later stage (Figure 4h; cluster-based permutation test, one-sided, $p = 0.013$; see Figures S14–15 for spatial maps; see Supplementary text for RSA analysis with node degree controlled), suggesting that the enhanced network representations in *HubToLeaf* cannot be attributed to node-degree information. In addition, analysis of neural geometry revealed a hub-attracted distortion in the *HubToLeaf* group (LMM2, two-sided, *HubToLeaf*, $\beta = -0.012$, $t(4760.00) = -3.27$, $p = 0.001$; *LeafToHub*, $\beta = 0.004$, $t(4760.00) = 1.19$, $p = 0.234$), consistent with predictions from models like the successor representation^{17,29}.

To localize these effects, we projected MEG data from the time window of interest into source space and performed a searchlight analysis (searchlight sphere: 10 mm radius; covering approximately 30 voxels) and repeated the RSA within each sphere in the source space. This revealed that the dorsal anterior cingulate cortex (ACC), with modest overlap in the supplementary motor area, significantly differentiated the groups in terms of network structure representation (Figure 4kl; LMM1, one-sided, FDR correction for whole-brain searchlight, peak $p = 0.004$, MNI $X = 0$, $Y = -5$, $Z = 45$). No significant differences were observed in hippocampal regions (Figure S17), and the effect was not explained by differences in univariate activity (Figure S16). Early network representations common to both groups were localized to the bilateral medial frontal cortex and right postcentral cortex (Figure S15, Table S1). Notably, to minimize movement during MEG recording, participants responded via button presses instead of continuous mouse tracking as used in Experiment 2. This less sensitive behavioral measure did not reveal significant performance differences between groups (Figure S12). Moreover, no significant correlations were observed between behavioral accuracy and neural representational strength (early cluster: *HubToLeaf*, $r = -0.38$, $p = 0.094$; *LeafToHub*, $r = 0.15$, $p = 0.51$; late cluster: *HubToLeaf*, $r =$

0.084, $p = 0.61$; *LeafToHub*, $r = 0.082$, $p = 0.73$), likely due to the limited sensitivity of the binary button-press accuracy measure.

In summary, MEG recordings provide convergent neural evidence that compressive learning facilitates the formation of inhomogeneous network structure representations in the human brain. These representations are localized primarily to the dorsal ACC, highlighting its potential role in structuring knowledge based on high-order network properties.

Two-stage computational modeling explains compressive learning

Finally, we developed two-stage computational models to elucidate the mechanisms of “compressive learning” observed in Experiment 2, which comprises two stages: Learning I (*HubToLeaf*, *LeafToHub*, *RandomWalk* pre-learning; 200 trials) and Learning II (*RandomWalk*; 800 trials).

During Learning I, we modeled the formation of initial network scaffold through the *HubToLeaf* and *LeafToHub* pre-learning paths (Figure 5a, left), based on four types of computational models – One-step model, Maximum Entropy model, Successor Representation model, and Hypergraph model (see details in the Supplementary Materials). During Learning II, image sequences generated via random walks were then incrementally integrated into the scaffold established in Learning I (Figure 5a, middle). Overall, the two-stage model captures how new experiences are anchored onto previously formed scaffold, ultimately giving rise to the acquisition of knowledge network (Figure 5a, right).

Specifically, the one-step (OS) model^{47,48} updates only direct transitions between consecutive items and ignores higher-order structure. The maximum entropy (ME) model³⁶ captures indirect relationships via probabilistic inference across multiple transitions. The successor representation (SR) model^{29,49,50}, rooted in reinforcement learning, captures extended temporal relationships by predicting future states discounted over time. The hypergraph model (HG) uses hyperedge concept (node and all of its neighbors; see more details in the Supplementary Materials) to represent substructure that could capture higher-order relationships among nodes, distinct from the pairwise representations of traditional graph theory. The four models vary in their treatment of higher-order network structure, with the OS model ignoring the higher-order properties, while the other three models are capable of learning higher-order properties to different extents. These structure-based

representations, either explicitly assumed in HG model or emerging from the ME and SR models, provide the foundation for “compressive learning”: hub-related substructures act as core skeletons of the network, scaffolding and facilitating subsequent random-walk learning in Learning II.

Model comparisons using the corrected Akaike Information Criterion (AICc) revealed a clear advantage for models incorporating higher-order structure, including the ME, SR, and HG models (Figure 5b). Specifically, the OS model performed significantly worse across all conditions, underscoring the critical role of higher-order relationships in network learning. Taken together, the modeling results reinforce the principle of compressive learning: pre-learning paths that highlight a network’s inhomogeneous higher-order structure (emphasizing hub-related substructures) promote the formation of a core scaffold, which in turn enables more efficient integration of subsequent information.

Discussion

We live in an era of information explosion, where data is overwhelming, fragmented, and unstructured—posing a significant challenge to humans’ innate drive for understanding. Conventional random-walk-based network learning operates through a bottom-up, blind sampling process that is inherently inefficient and time-consuming. Here, we introduce an approach—compressive learning—which embeds core structural properties (specifically, node-degree inhomogeneity) into a pre-learning trajectory to enhance the efficiency of network learning. This conceptual framework is supported by two large-scale behavioral experiments and is further validated through neural evidence and computational modeling. Together, our findings demonstrate that humans are not only sensitive to higher-order structural properties during network learning, but can also exploit them more effectively when these properties are presented in a structured manner that mirrors the organization of the underlying network.

Our findings are consistent with prior work demonstrating that learning paths critically shape network acquisition^{41,51}. While Qian et al.⁴¹ emphasized the need to reinforce leaf nodes in heterogeneous networks—a conclusion that may appear to contrast with our hub-focused results—both perspectives highlight the central role of hubs in structuring networks, albeit with different goals. Specifically, our study aimed to rapidly establish scaffolds of inhomogeneous networks by prioritizing hub-related connections, whereas Qian et al.⁴¹ focused on compensating for memory

degradation by strengthening leaf-related connections. Crucially, our work goes beyond previous studies by introducing a graph-based perspective, controlling for path-related confounds by testing identical sequences, and directing attention to inhomogeneity—an ecologically salient but underexplored property of real-world networks. Notably, although participants were not informed about the underlying graph structure, they were explicitly instructed to predict the next image based on learned pairwise transitions. This differs from previous implicit network learning paradigms^{15,29,36}. Implicit learning reflects the gradual accumulation of statistical regularities, whereas explicit learning involves deliberate encoding of pairwise relations and greater demands on working memory and strategic processes^{52–56}.

The central concept of “compressive” here has two meanings: it describes both the structural property of networks and the nature of the learning process. Inhomogeneous networks such as scale-free networks are structurally compressible, as a few high-degree hubs form a skeleton that efficiently captures global organization (Experiment 1). A learning path is compressive such that it introduces the structural skeleton first, providing a condensed schema to support later acquisition (Experiment 2&3). In fact, the name of “compressive learning” draws inspiration from compressive sensing^{34,35}, where high-dimensional signals are reduced to a few critical components for efficient reconstruction. Analogously, compressive learning decomposes abstract network structures into substructures of varying significance—ranging from hub-related to leaf-related elements—based on node-degree inhomogeneity. Hub-related substructures, with their dense connectivity, provide more global information about the network and are prioritized in the *HubToLeaf* pre-learning path. This ordering is hypothesized to yield a compact structural skeleton in the brain that facilitates subsequent integration of new information. Conversely, the *LeafToHub* path emphasizes lower-information substructures early on, potentially resulting in a less efficient scaffold.

Humans are naturally inclined toward statistical learning^{57,58}—the ability to track associations between sequentially presented items over time. Ideally, these local, pairwise associations could coalesce into a coherent network representation. However, local learning is often limited by constraints in memory and attention^{3,59,60}, leading to fragile and error-prone associations. In contrast, higher-order network properties—such as community structure and node-degree inhomogeneity—capture non-local regularities that facilitate a more global understanding of the network^{14,33}. While such representations may abstract away finer local details^{26,61}, they provide an

efficient scaffold for integrating new information and generalizing beyond observed data. These higher-order features are also pervasive in real-world networks^{30,62–65}, further underscoring their relevance in efficient information encoding. The SVD-based quantitative analysis supports that networks with stronger inhomogeneity are more compressible, suggesting that such structural asymmetries are particularly informative. Crucially, we provide empirical evidence that node-degree inhomogeneity enhances learnability and can be leveraged to design more effective pre-learning sequences.

The prefrontal cortex has been consistently implicated in sequence memory and the temporal organization of experience^{66–69}, as well as in higher-order structure learning⁷⁰. In particular, the medial prefrontal cortex (mPFC), which is closely related to the dorsal ACC (dACC) both anatomically and functionally⁷¹, has been shown to support the formation of abstract structural and relational models^{15,18,72}. Here, early neural representations—common across both *HubToLeaf* and *LeafToHub* groups—emerge in bilateral superior frontal and right inferior parietal cortices, consistent with prior findings⁷³. Importantly, selective engagement of the dACC following *HubToLeaf* pre-learning may reflect the emergence of a compact predictive representation of the network that supports efficient sequence inference and decision guidance. The ACC is known to support high-level cognitive functions such as hierarchical decision-making^{73–76}, foraging⁷⁷, learning^{78,79}, and context encoding⁸⁰. Our results thus extend its functional repertoire to the formation and application of structured network representations, which situates “compressive learning” within a broader literature implicating frontoparietal circuits in relational abstraction and model-based control⁸¹. Importantly, these ACC effects are not attributable to differences in overall activation, node salience, or hub-related substructure. Instead, the ACC appears to track the structural scaffold consistent with the underlying network topology, highlighting the neural correlates of network structure learning.

We proposed a two-stage computational model to elucidate the underlying computational basis. Model comparisons revealed that frameworks incorporating higher-order relational structure consistently outperformed those based on simpler transition statistics. This network skeleton formed during Learning I captures structural compression, enabling learners to integrate new nodes through higher-order patterns during Learning II. Importantly, although the advantage of compressive learning builds on priming from the pre-learning *HubToLeaf* path, it does not simply reflect a primacy effect on specific pairs. Rather, it highlights the essential role of hub-related

associations in shaping the network scaffold that extends across the entire structure. Moreover, the learning advantage could not be attributed to early repetition in the *HubToLeaf* path, as the hub-related edges—although sharing the same hub node—were distinct from one another.

From a broader perspective, the two-stage learning framework—involving scaffold formation (Learning I) followed by structure integration (Learning II)—offers a generalizable model for learning from complex, nonlocal input spaces. This process is distinct from curriculum learning^{4,82}, which typically utilizes generalization from easy to hard. In contrast, the *HubToLeaf* strategy during Learning I follows a hard-to-easy trajectory (Figure S7A), consistent with statistical learning principles and successor representation predictions that transitions with lower conditional probability are typically learned less well than those with higher probability^{49,83–85} (see also Figure S7B). Moreover, unlike previous studies that emphasize temporal statistical properties of input sequences^{33,36}, our approach explicitly embeds higher-order relationships into pre-learning paths. Future work could integrate and prioritize higher-order structures in the training procedures to facilitate learning efficiency of artificial intelligent agents⁸⁶.

Finally, our findings resonate with the literature on memory schemas, which proposes that structured knowledge frameworks facilitate the rapid integration of new information^{9,11,87}. Schema theories suggest that learning is accelerated when incoming information can be embedded within an existing relational scaffold, a process often associated with coordinated hippocampal–prefrontal interactions⁸⁸. Within this perspective, compressive learning may reflect the emergence of a compact structural representation that summarizes the higher-order organization of a relational network. HubToLeaf pre-learning may provide an efficient route to construct such a scaffold by first capturing global structural relationships, enabling subsequent learning to proceed through inference within a compressed representation rather than independent pairwise encoding. More broadly, such structural scaffolds may constitute a key component of internal mental or world models that allow humans to organize complex information and perform flexible inference under limited experience.

Methods

Subjects. The study had been approved by the Institutional Review Board of School of Psychological and Cognitive Sciences at Peking University (#2019-02-08). Subjects in all experiments, both online and in the laboratory, provided written informed consent in accordance with the Declaration of Helsinki.

Experiment 1. For the in-laboratory experiment, 20 human subjects (aged 18-30, 50 female, 30 male) of each group were recruited from Peking University. All of them had normal or corrected-to-normal vision. The behavioural experiment lasted for 60 min, and subjects received 55+(0~30) RMB for their time (basic participation fee + reward). For each group in the online experiment, 20 human subjects (aged 18-30, 39 female, 39 male, 2 unknown) were recruited from the Prolific platform. The online experiment was decomposed into two sessions. Online subjects were allowed to participate in the 1st session if they were university students, aged between 18 and 30 years, fluent in English, and had an approval rate higher than 90. In the first session, subjects read the instructions and then completed a short examination to prove that they had fully understood the task. Only those subjects who passed the test within 5 attempts in the 1st session were invited to the 2nd session (the main experiment). The average completion time for the 1st and 2nd sessions was ~10 and ~50 min separately. In addition to the basic participation fee (£7.25), subjects had the opportunity to obtain an additional bonus (max. £4), which was accumulated across trials depending on their accuracy and RTs.

Experiment 2. 78, 80, and 80 subjects (aged 18-30, 114 female, 105 male, 19 non-binary or unknown) from the Prolific platform were recruited for the *RandomWalk*, *HubToLeaf*, and *LeafToHub* groups in Experiment 2, respectively. Subjects who had participated in Experiment 1 were disqualified. The same compensation scheme used for online subjects in Experiment 1 was applied. For the in-laboratory experiment, 21 human subjects (aged 18-30, 35 female, 28 male) for each of the path groups were recruited from Peking University. In-lab participants were compensated for their time (¥60 per hour) with performance dependent bonus (max. ¥20).

Experiment 3. 40 subjects (aged 18-30, 19 female, 21 male) were recruited from Peking University for the MEG experiment. Participants were compensated for their time (¥100 per hour) with performance dependent bonus (max. ¥30).

Across all experiments, gender information was collected based on subjects' self-report. Gender was not considered as a factor in the study design, and no gender-based analyses were conducted because the study was not designed or powered to test gender differences.

Apparatus and stimuli

In-laboratory subjects were seated approximately 86 cm in front of a 21.5-inch iMac monitor (47.9×26.9 cm, 2,048×1,152 pixels, 60-Hz refresh rate). The display of stimuli and the recording of responses were controlled by the iMac computer using MATLAB R2017a and PsychToolbox-3^{89,90}. The visual angle of each image was 2.25°, and all images in a display were equidistantly

distributed on an invisible circle that subtended a visual angle of 9° . For the online experiment, the stimuli were presented in the unit of height, defined as a relative measure to the height of the window. Thus, the physical sizes and locations of the stimuli would change with different monitors. The stimuli were 16 cartoon images with abstract semantics (<https://www.flaticon.com/packs>).

Transition network

Unbeknownst to the subjects, we used four different transition networks (Figure 2a) in Experiment 1 to generate the image sequences across trials, with each node denoting one image in the image set (Figure 2a, left). The four networks were matched in their basic topological features, such as the same number of nodes (16) and edges (32), and an average node-degree (number of edges that are directly connected to the node) of 4. However, their node-degree distributions differed, resulting in diverse heterogeneity (Figure 1a). The Lattice network was a spherical shape, with each node fully connected to its adjacent four nodes. The Random network was generated based on the Erdős-Rényi model by randomly connecting each pair of nodes with the probability p , and the corresponding average degree of a random network was given by $p(n - 1)$, where n is the total number of nodes⁹¹. The Small-world network was constructed by randomly rewiring the edges on regular (or Lattice) networks with the probability $p = 0.3$ ⁹². In the current setting, the node-degree distributions of Random and Small-world networks were similar to quasi-normal. The Scale-free network was constructed with the Barabási-Albert model⁹³, which possessed a unique feature of high connectivity at very few nodes (the hubs). All subjects experienced the same 16 images and the same transition network within a group, but the correspondence between nodes and images was shuffled across subjects. Based on the objective transition network, four corresponding minimal distance matrices (MDM) were constructed, with each cell denoting the minimal transition steps between two nodes.

Probabilistic sequential prediction task

Each trial started with a cue image, followed by the presentation of target and distractor images (Image onset). The target image was directly connected to the cue image in the transition network (one-step transition), while the distractor image was randomly selected from nodes that were more than one step away from the cue. In each trial, only one target image and one distractor image were displayed. Subjects' task was to click the target image using the mouse as soon as possible after the image onset without sacrifice of accuracy, during which their time-resolved trajectories were recorded (Figure 2b).

Experimental procedure

Each trial started with the cue image on the screen for 0.7 s (except for the 1st trial of each block, which was presented for 2 s), after which one target and one distractor image were presented for 2 s (Image onset). The subjects' task was to select the target image as accurately and quickly as possible through learning and feedback.

To facilitate the network learning, after 0.8 s presentations of alternative images, a small gray square (the hint) appeared gradually on the target image to indicate the correct answer. Note that subjects were encouraged to make a response before hint onset if they were confident. Once subjects made responses, auditory feedback was presented indicating whether the response was correct, incorrect, or time-out. The feedback screen lasted for 0.7 s.

Subjects received rewards on each trial based on their behavioral performances. Specifically, a correct response before hint would yield 100 virtual coins. Otherwise, the accessible coins decreased linearly with time, declining from 100 to 0 within 0.5 s and from 0 to -50 for the remaining 0.7 s. In other words, before- and after-hint responses yielded higher and lower rewards, respectively. Initially, subjects must rely on the hint, which appeared on the target image with the same gray scale as the background at 0.8 s and gradually increased to the largest gray scale within the next 0.8 s, to make a correct response. Otherwise, time-out or incorrect selection would lead to a loss of 50 coins loss. In the laboratory experiment, in order to control the mouse's moving speed, subjects were informed that the cursor's position would be forced back to the screen center if their moving speeds were too fast.

Generation of priming scenarios in Experiments 2 and 3

In Experiment 1, the target image in each trial served as the cue image in the next trial for subjects to make predictions, with the target image sequences being prescribed by a random walk on the transition network. A random walk is a path across a transition network created by taking repeated random steps between any two directly connected nodes³¹. In contrast, in Experiments 2 and 3, the continuity of the cue between target images across trials was broken for the first 200 trials. Specifically, the cue-target-distractor combination in each trial was ranked primarily based on the node-degree of cue image, and then ranked again based on the node-degree of target image under the same cue image (Figures 1f and 3b). Hence, the first 200 trials were sorted in descending order based on the node-degree of both the cue and target images for the HubToLeaf group. Conversely, in the LeafToHub group, the trials were sorted in ascending order. In summary, the three groups (Random, HubToLeaf, LeafToHub) differed only in their trial organisations in the first 200 trials (priming scenarios), while the stimulus exposure in the remaining 800 (Experiment 2) or 600 trials (Experiment 3) was similar. We therefore referred to the first 200 trials as the Learning I phase, and the remaining trials with random walk transitions as the Learning II phase. Subjects were not informed about the priming scenarios; they were only told that the cue image would be repeated over a period.

Time-resolved mouse trajectory analysis

In the laboratory experiment, we recorded the mouse trajectory at a sampling rate of 60 Hz during each trial for each subject. The trajectory initiating from the center point and ending at the target or distractor image could reflect subjects' internal decision process. For the online experiment, we adjusted the sampling rate to 60 Hz by interpolation due to the different operating systems used by the subjects. At each sampling point within a trial, we calculated the absolute angle between

the cursor dwell position (pixel unit for the in-laboratory experiment and height unit for the online experiment) and each alternative image (target or distractor). A smaller absolute angle indicates a stronger decision preference towards a specific image.

We calculated the intersection angle between each sample point of the mouse trajectory and the center of the target or distractor images within a trial, with the center of the screen as the origin. The resulting mouse-origin-target angle (θ_t) and mouse-origin-distractor angle (θ_d) were then imported into a binary classifier to determine subjects' internal decision at the current sample point (Figure 2c). Specifically, a correct choice was exported if the mouse's dwell position was closer to the target, i.e., $|\theta_t| < |\theta_d|$; otherwise, an incorrect response was output.

Time-resolved accuracy computation. Time-resolved accuracy was computed to examine the temporal dynamics of decision-making. The accuracy at time point t , $Accuracy(t)$, was calculated as the proportion of trials classified as correct (choosing the target) across all available trials at that specific time point. Time-resolved accuracy served primarily as a validation measure to confirm that participants could make above-chance predictions before hint onset across all experimental conditions.

Overall accuracy computation. To quantify learning performance for between-group comparisons while avoiding hint-related biases, we computed overall accuracy using the same binary classification framework but extracting a single time point per trial. Specifically, we extracted the classification result at the moment of mouse click for response-before-hint trials, or at the position immediately before hint onset (0.8 s) for response-after-hint trials (Figure S2). This approach ensured that each trial contributed equally to the accuracy measure regardless of reaction time, making it suitable for statistical comparisons between experimental conditions while maintaining the same underlying classification logic as the time-resolved analysis.

The key technical distinction is that time-resolved accuracy aggregates across time points within trials (resulting in variable trial contributions over time), while overall accuracy aggregates across trials using a single decision point per trial (ensuring equal weighting regardless of reaction time).

The 16×16 accuracy matrix reported in Figure 3f was derived from overall accuracy, calculated as the average accuracy across trials in Learning II (testing phase) for each (one-step) cue-target pair. Hub-related accuracy was computed from trials involving Hub nodes (degree > 7), whereas Leaf-related accuracy was computed from trials involving only Leaf nodes (degree ≤ 7).

MEG Experiment

Apparatus and MEG acquisition. Visual stimuli were presented with a projector (Panasonic PT-DS12KE) at a refresh rate of 60Hz. Auditory stimuli were presented with a pair of earphones. Subjects were seated in a magnetically shielded room and reacted with a MEG-compatible response box. Neuromagnetic signals were recorded at 1,000 Hz by a 306-channel system (Elekta-Neuromag) with 102 magnetometers and 204 gradiometers while subjects were seated 1 m in front

of a projection screen and performing the task. MRI T1 data were acquired after MEG recording, using an MP-RAGE sequence on a Siemens 3T TRIO scanner, with voxel resolution of $1 \times 1 \times 1 \text{ mm}^3$ on a $176 \times 192 \times 192$ grid (echo time = 2.98 ms, inversion time = 1,100 ms, repetition time = 2,530 ms). Eye movement was monitored online by an eye tracker (Eyelink 1000 plus, remote mount) at 1000 Hz. Five-point calibration was performed every 100 trials.

MEG preprocessing. The preprocessing of MEG data was implemented with MNE-Python (version 1.3.1)⁹⁴, following the FLUX pipeline⁹⁵. Maxwell filter and Signal-space separation were performed first to remove outside-helmet artifacts. The data were then high-pass filtered at 0.5 Hz and low-pass filtered at 40 Hz, followed by downsampling to 250 Hz. Independent component analysis was then performed on the filtered data. Ocular and cardiac components were manually identified for each subject and then subtracted from the data. All epochs were aligned at the onset of cue image. Only trials with correct responses were included for the following MEG analysis.

Representational similarity analysis (RSA)

The general principle of RSA is comparing the representational similarity between the neural activity and a proposed model⁹⁶.

Sensor-level RSA. Neural representational dissimilarity matrices (RDMs) were constructed using whole-brain sensor data as multivariate neural patterns at each time point^{17,43–46}. We recorded multivariate neural responses for each of the 16 cue images and computed neural dissimilarities for every image pair to construct a 16×16 neural RDM, $N(t)$, at each time point. To obtain robust estimates, we performed a resampling procedure to balance trial numbers across different node image. For each of 100 iterations, we randomly drew 50 trials with replacement from trials with the same cue image and averaged the MEG data across these trials. This procedure was repeated for each of the 16 cue nodes. At each time point, we calculated the distance ($1 - \text{Pearson's } r$) between the topographic maps of different nodes, then averaged RDMs across the 100 iterations. To denoise, the RDM time series were baseline-corrected by subtracting the average RDM over a baseline window (from cue onset to 500 ms before target onset).

The model dissimilarity matrix (MDM), M , characterized the underlying network structure as a 16×16 matrix where each element represented the shortest path distance between two nodes according to the network graph.

To examine how the representation of network structure changes over time, we analyzed the relationship between neural RDM and MDM at each time point. LMM1 were specified to test whether the subjects represent the network structure and whether group difference exists in representation strength. Cluster-based permutation test was used for correction (see Supplementary Text for details).

For visualization, neural RDMs and MDM are normalized and then projected to 2-dimensional Euclidean space with multidimensional scaling. Multidimensional scaling tries to reconstruct the coordinates of all nodes on a Euclidean map that minimize the summed absolute

difference between the distance in dissimilarity matrices and the reconstructed distance. Multidimensional scaling is implemented with Python package sklearn⁹⁷.

Leaf-related and hub-related submatrices. In the Scale-free network, nodes with more than 7 connections are defined as hub nodes, and the rest nodes are defined as leaf nodes. In Experiment 3, to examine whether the specified path influences the entire network, we fitted LMM1 separately on hub-related and leaf-related submatrices. Both submatrices are subsets of a full dissimilarity matrix. Leaf-related submatrix contains only distances between leaf nodes. Hub-related submatrix is the complement set of leaf-related submatrix, which contains hub distances to both hub and leaf nodes.

Source reconstruction. Cortical reconstruction from T1-weighted MRI was performed using FreeSurfer (version 7.3.2). MEG-MRI co-registration, source and forward model computation were implemented using MNE-Python (version 1.3.1). The baseline covariance matrices of the sensors were computed with a time window from -0.5 s to 0 relative to cue onset. Linearly constrained minimum variance beamforming⁹⁸ was used to project epoched data from both magnetometers and gradiometers onto a volume bounded by each subject's brain with a regularization coefficient of 0.05. Voxels were evenly spaced 5 mm apart. To make group-level inference, we morphed all source-level data into a common source space⁹⁹ before statistical analysis.

Source-level RSA searchlight. We performed source-level RSA searchlight analysis to identify cortical regions responsible for network representation. Sensor-level signals within the two significant time windows (50-ms width, centered at 370 ms and 1196 ms) were projected to source space and analyzed separately. For each voxel, we included all voxels within a 10 mm radius sphere, which contained an average of approximately 30 voxels per sphere (5th percentile: 18; 25th percentile: 29; 75th percentile: 32), providing adequate sample sizes for reliable correlation analysis. We applied the same RSA procedure to the voxels as performed at sensor level, with two modifications to reduce computational cost. First, we reduced the number of resampling iterations to 25. Second, we applied false discovery rate correction instead of cluster-based permutation testing for multiple comparison correction.

Notably, performing multivariate RSA analysis simultaneously in both time and space with multiple comparison correction is quite challenging in MEG studies. Instead, a two-stage strategy — first identifying time windows from global sensor-level dynamics, then localizing sources via searchlight within the time window — has been widely used in the MEG RSA field^{100,101}. Here we followed the same procedure.

Node-degree regression analysis

At each time point and each channel, MEG data were baselined and then regressed with node-degree of cue image.

Statistical Analysis

Linear mixed models (LMMs) and generalized linear mixed models (GLMMs). LMMs and GLMMs were implemented for group-level inference to account for both group-level and participant-level variance and to control nuisance variables. Both variables of interest and variables for control are specified as fixed effects. Specification of random effects followed the keep-it-maximal principle to prevent false positive results^{102,103}. Random effects and their correlations would be reduced gradually in case of singularity. Recency of nodes, node-degree of target nodes, and minimal distance of distractor nodes are controlled as fixed and random effects. While LMM assumes Gaussian distribution for noise, GLMM assumes binomial response noise. The models are fitted with maximum likelihood estimation. The significance of each fixed effect is determined by Welch-Satterthwaite t test for LMM and likelihood ratio test for GLMM. The Bonferroni correction is used in post-hoc comparison. Cluster-based permutation test is used for test in the RSA time course¹⁰⁴.

Cluster-based permutation test for multiple comparison correction. The massive statistical test at each time point might cause excessive type I error. We performed a cluster-based permutation test to identify significant temporal clusters¹⁰⁴, which was widely accepted^{105–107}. The general intuition of the correction is that, instead of testing whether each time point survives the statistical test, experimenters can test whether each cluster survives the test. The statistics we used to characterize effects here were derived from LMM1. Accordingly, for each permutation, we fit the same LMM1 as we did in the original analysis, except that we shuffled the node identities in the minimal distance matrix and subjects' group labels. We then performed the same Satterthwaite t -test at each time point. Consecutive points that survived thresholding ($p < 0.05$) were clustered. The maximum summed t value among all clusters within each permutation was kept. Multiple permutations (1000 times) were performed and resulted in a surrogation distribution of max summed t values. Summed t values from original empirical data were looked up in the surrogation distribution for a p value. Null hypothesis was rejected if any cluster from the unshuffled data was larger than 95% of summed t in the surrogate distribution.

In node-degree analysis, since channel dimension was not collapsed as in RSA, significant cluster was determined jointly by both temporal and channel dimension.

Network heterogeneity analysis

Typically, the degree heterogeneity¹⁰⁸ of a network is defined as

$$\xi = \frac{\sum_k k^2 p(k)}{[\sum_k k p(k)]^2} = \frac{\langle k^2 \rangle}{\langle k \rangle^2}, \quad (1)$$

where $p(k)$ is the distribution probability of degree k in the network.

Network compressibility analysis

Many natural or technological systems are often characterized by a low-rank nature. As demonstrated in ref.¹⁰⁹, using singular valued decomposition (SVD), the low-rank characteristic of *Drosophila melanogaster*'s hemibrain connectome can be experimentally verified. According to

the Schmidt–Eckart–Young–Mirsky theorem¹¹⁰, SVD provides the optimal low-rank approximation of a matrix in terms of the Frobenius norm. Therefore, we can use SVD to measure how the network structure is captured by a low-dimensional structure. For any adjacency matrix $\mathbf{A} \in R^{n \times n}$, we can represent it as

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \quad (2)$$

where $*$ denotes the complex conjugate transpose and $\mathbf{U}, \mathbf{V} \in C^{n \times n}$ are unitary matrices whose conjugate transpose is equal to its inverse, namely $\mathbf{U}^*\mathbf{U} = \mathbf{V}^*\mathbf{V} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix. The matrix $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n) \in R^{n \times n}$ is a diagonal matrix of singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n \geq 0$. According to Eq. (2), matrix $\mathbf{A}$ can also be given by the sum of rank-one matrices, weighted by singular values

$$\mathbf{A} = \sum_{i=1}^n \sigma_i \mathbf{u}_i \mathbf{v}_i^*, \quad (3)$$

where $\mathbf{u}_i$ and $\mathbf{v}_i$ are the i th column vector of matrices $\mathbf{U}$ and $\mathbf{V}$, respectively. In light of this, SVD offers a hierarchy of low-rank approximations by truncating the sum to include only the leading singular values. Correspondingly, if a matrix is reconstructed more accurately under the same rank approximation, it is considered to be more compressible. We quantify the compressibility of a network using the reconstruction error, here defined as the mean squared error (MSE) between the adjacency matrix of the network and its low-rank approximation across all possible truncation levels, given by

$$\varepsilon = \frac{1}{n} \sum_{i=1}^n \left\| \mathbf{A} - \sum_{j=1}^i \sigma_j \mathbf{u}_j \mathbf{v}_j^* \right\|_F^2, \quad (4)$$

where $\|\cdot\|_F$ denotes the Frobenius norm.

Data Availability

Source data for Figures 1-5 and Supplementary Figures and Tables produced in this study are available at <https://doi.org/10.17605/OSF.IO/DM3Z9>¹¹¹.

Code Availability

All codes from this study are available at <https://doi.org/10.17605/OSF.IO/DM3Z9>¹¹¹.

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Author Contributions Statement

X.R., A.L., and H.L. originally conceived the study. X.R. and M.W. performed the experiments. X.R., M.W., and T.Q. analysed the data. T.Q., A.L., and H.L. performed theoretical analyses. X.R., M.W., T.Q., F.F., A.L., and H.L. wrote the paper.

Competing Interests Statement

The authors declare no competing interests.

Figure Legends

Figure 1. The schematic diagram of compressive learning. **a**, Sixteen images (left) are randomly assigned as nodes in an underlying network (right). Participants were asked to learn the network through a pair-by-pair learning path (bottom). **b**, Four underlying networks with increasing inhomogeneity are designed. Up: Lattice (pink), Random (green), Small-world (blue), Scale-free (orange) networks. Node-degree is denoted by node size and brightness. Bottom: the distribution of node-degree of the four networks. **c**, Compressibility is measured by how less compression loss the compressed network will have after SVD compression. We measure the compression loss with the area under the mean squared error (MSE) curve (see Eq. (4) in Methods). The smaller the area, the higher compressibility. Scale-free is the most compressible network among the four. **d**, Compressibility increases with node-degree inhomogeneity, indicated by decreasing compression loss. Networks of different inhomogeneity levels, as quantified by Eq. (1), are generated from multiple network generation models, denoted by different colors and shapes. **e**, Schematic prediction. As the network compressibility along with inhomogeneity increases, participants learn the network better. **f**, Schematic diagram of compressive learning: Designed pre-learning path primes the inhomogeneous structure of a network, builds a network skeleton, and thus, facilitates the subsequent random-walk learning. The black-and-white icons in the image set shown in figure 1a were obtained from Flaticon. In the 4×4 image set, icon positions are specified as (row, column), with rows indexed from top to bottom and columns from left to right. Icons at positions (1,1), (1,2), (1,3), (1,4), (2,1), (3,4), and (4,1)) were created by bqlqn; icons at positions (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (4,2) and (4,3) were created by Freepik; and the icon at position (4,4) was created by edt.im. See Supplementary figure S21 for a detailed attribution of all icons used in the experiments.

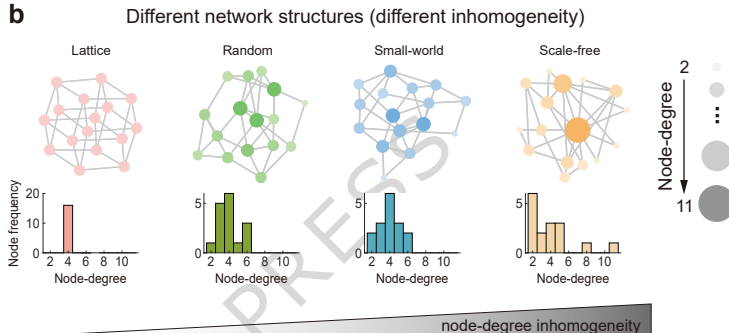
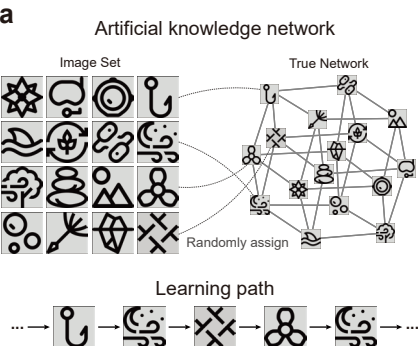
Figure 2. Network learning paradigm and behavioural performance (Experiment 1). **a**, Four groups learned four types of 16-node, undirected and unweighted transitional networks: Lattice (pink), Random (green), Small-world (blue), Scale-free (orange) networks with matched basic properties. Node-degree is denoted by node size and brightness. The same 16 black-and-white icons, created by bqlqn, Freepik and edt.im, were randomly assigned to the network nodes per subject. **b**, Sequential prediction task. Left: after cue image (Cue), subjects predict its succeeding image by mouse-clicking one of the two images (Target-Distractor). After 0.8 s, a gray square gradually appears on the target as a hint (Hint visible). Response before hint is encouraged and yields higher rewards. Middle: current-trial target image becomes the next-trial cue (dashed arrows). Right: target image sequence is generated via random-walk within corresponding network. Light green to dark blue arrows denote progression across trials. **c**, Time-resolved mouse trajectory analysis. Left: at each time point, angle between mouse position and target or distractor images is computed as $\theta_T(t)$ and $\theta_D(t)$, respectively. Smaller $\theta_T(t)$ than $\theta_D(t)$ indicates correct prediction and larger $\theta_T(t)$ than $\theta_D(t)$ indicates incorrect prediction. Accuracy ($Accuracy(t)$) is calculated as the proportion of correct classifications. Right: overall prediction accuracy at decision point. **d**, Time-resolved prediction accuracy of four networks. Gray and colored lines denote individual subjects ($N=40$ per network) and grand averages. Shading areas indicate the S.E.M. Vertical dotted line indicates hint onset. Colored dots on top indicate group-level significance (compared to 0.5 chance level; one-sample t -test, two-sided, $p < 0.05$, FDR correction across time). **e**, Overall prediction accuracy. Overall accuracy is computed by extracting one classification result per trial from the same trajectory analysis (before hint onset), then averaged across trials for each subject ($N=40$ per network). In the violin plots, white circles indicate the median, boxes represent the interquartile range (25th-75th percentiles), and whiskers extend to the minima and maxima within 1.5 times the interquartile range from the lower and upper quartiles. Colored stars: above-chance accuracy within a group (compared to chance-level 0.5; one-sample t -test, two-sided). Black asterisks: post-hoc comparison with Bonferroni correction between groups (GLMM1, ***: $p < 0.001$).

Figure 3. Compressive learning facilitates subsequent learning performance (Experiment 2). **a**, Left: scale-free network with Hub nodes (more connections) and Leaf nodes (fewer connections). Middle: three pre-learning paths – *RandomWalk* (yellow), *HubToLeaf* (purple), *LeafToHub* (green). Right: during Learning I (200 trials), groups learned sequences generated by different pre-learning paths. During Learning II (800 trials), they learned identical random-walk image sequences. Only Learning II performance was compared between groups. **b**, Generation of three pre-learning paths (Learning I). *RandomWalk* was generated by a 200-step random walk. *HubToLeaf* and *LeafToHub* paths were created by reordering the same sequence according to the node degree of cue and target images in descending and ascending order, respectively. All paths contain the same black-and-white icons (created by bqlqn and Freepik) with identical frequencies but different orders. **c**, Time-resolved prediction accuracy (Learning II). Gray and colored lines denote individual subjects and grand averages (mean±S.E.M.). Vertical dotted line indicates hint onset. Colored dots indicate group-level significance (two-sided one-sample t -test, $p < 0.05$, FDR correction). **d**, Temporal evolution of prediction accuracy (Learning II). Triangle: *HubToLeaf* vs. *Random*; diamond: *HubToLeaf* vs. *LeafToHub*. Filled, $p < 0.01$; hollow: $0.01 \leq p < 0.05$ (GLMM2 and Bonferroni corrected post-hoc comparisons in each time bin; FDR correction across bins). **e**, Overall prediction accuracy (Learning II). *: $p = 0.017$; ***: $p < 0.001$ (GLMM2 with Bonferroni corrected post-hoc comparisons). **f**, Overall accuracy for Hub-related and Leaf-related pairs (Learning II). Heatmaps show the mean accuracy for each cue–target pair across trials. Red-highlighted entries indicate Hub-related transitions, whereas other valid entries represent Leaf-related pairs. Accuracy is shown in grayscale, with darker shades indicating higher accuracy; white cells denote invalid cue–target transitions. Violin plots summarize the average accuracy for Hub-related and Leaf-related transitions across groups (statistics as in Fig. 3e). ***: $p < 0.001$; n.s.: $p \geq 0.05$. For **d**, **e** and **f**, the sample sizes for the three groups were 78, 80, and 80, respectively. In the violin plots, white circles indicate the median, boxes represent the interquartile range (25th-75th percentiles), and whiskers extend to the minima and maxima within 1.5 times the interquartile range.

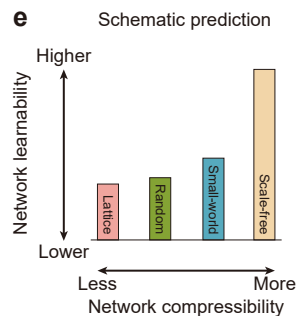
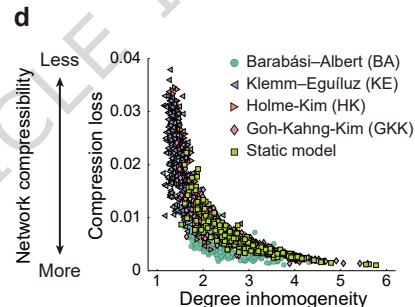
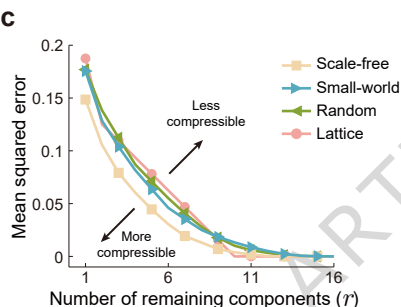
Figure 4. Compressive learning enhances network structure formation in dorsal ACC (Experiment 3, MEG). **a**, Two groups participated in Experiment 3: Learning I (200 trials; *HubToLeaf* or *LeafToHub* paths) and Learning II (600 trials; *RandomWalk* path). Only Learning II is analyzed. **b**, Each trial is labelled by the identity of cue image within network labels each trial. Multivariate neural dissimilarity matrix (neural RDMs) are calculated for all node pairs at each time point, resulting in time-resolved neural representational dissimilarity. **c**, Model RDM is constructed based on the corresponding minimal transitional distance within the network. Regressing neural RDM by model RDM, yielding neural representational strength of network structure. **d**, Time-resolved neural representational strength of network structure during Learning II, for *HubToLeaf* (purple) and *LeafToHub* (green) groups (p values from LMM1 with cluster-based permutation test; one-sided). Time window with significant group differences (shades in orange) is used as the time-of-interest for further analysis. **e**, Two-dimensional representation of the ground-truth network structure (multidimensional scaling of model RDM; MDS). Dots denote images within network. Dot size indexes node-degree. **f, g**, MDS results (left) and neural RDMs (right) within the time-of-interest for *HubToLeaf* and *LeafToHub* groups. **h**, Time-resolved neural correlation of node-degree (p value from independent t -test with cluster-based permutation, two-sided). **i**, Neural representational strength of sub-network structures – Hub-related and Leaf-related submatrix – within the time-of-interest, for *HubToLeaf* (purple) and *LeafToHub* (green) groups. (p values from interaction effects of LMM1, one-sided, uncorrected) **j**, Average neural distances of each node decrease with node degree in the *HubToLeaf* group (left, $N=20$), but not in the *LeafToHub* group (right, $N=20$; Slope coefficients β and p values from LMM2; two-sided). **k**, Source-level analysis. Sensor-level data is projected into source space to perform the same RSA analysis within spheres of 10-mm radius. **l**, Searchlight results for *HubToLeaf-LeafToHub* group difference on neural representational strength of network structure (Whole-brain t -value map of group effect, LMM1, one-sided; Colored area, $p < 0.05$, FDR corrected; peak $p = 0.004$; Brown contour, cingulate cortex; Yellow contour, supplementary motor area). Shadings in **d** and error bars in **i-j** indicate standard error, whereas shadings in **h** denotes S.E.M.

Figure 5. Two-stage computational model. **a**, Illustration of two-stage computational model. During Learning I (pre-learning path), *HubToLeaf* (upper, purple) and *LeafToHub* (lower, green) groups learned hub-related and leaf-related substructures in reversed order. During Learning II (black), both groups learned identical random-walk image sequences (black) based on network scaffold formed during Learning I, resulting in finally learned relational network for *HubToLeaf* (upper, purple) and *LeafToHub* (lower, green) groups. **b**, Model comparisons among the one-step (OS), maximum Entropy (ME), successor representation (SR), and hypergraph (HG) models for *HubToLeaf* ($N=80$), *LeafToHub* ($N=80$), and *RandomWalk* ($N=78$) groups. Lower $\Delta AICc$ values indicate better model fit. Error bars represent mean $\pm$ S.E.M. Notably, the four models vary in their treatment of higher-order network structure, with the OS model ignoring higher-order properties, while the other three models learn higher-order properties based on different principles (see details in Supplementary Materials). The direct comparison of AICc between the OS model and the three models incorporating higher-order structural properties (ME, SR, and HG) yields an average difference greater than 9^{112} , indicating a clear advantage for models incorporating higher-order structure.

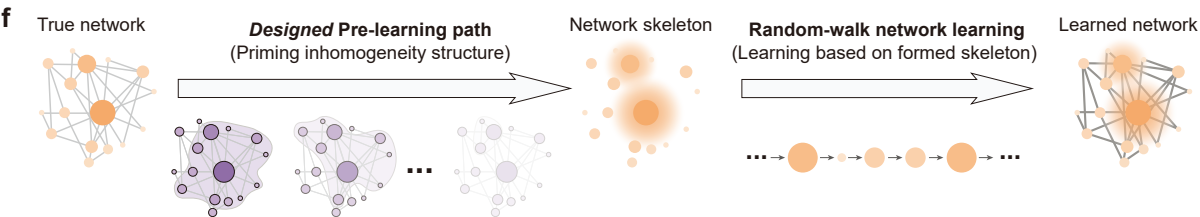
Knowledge Network

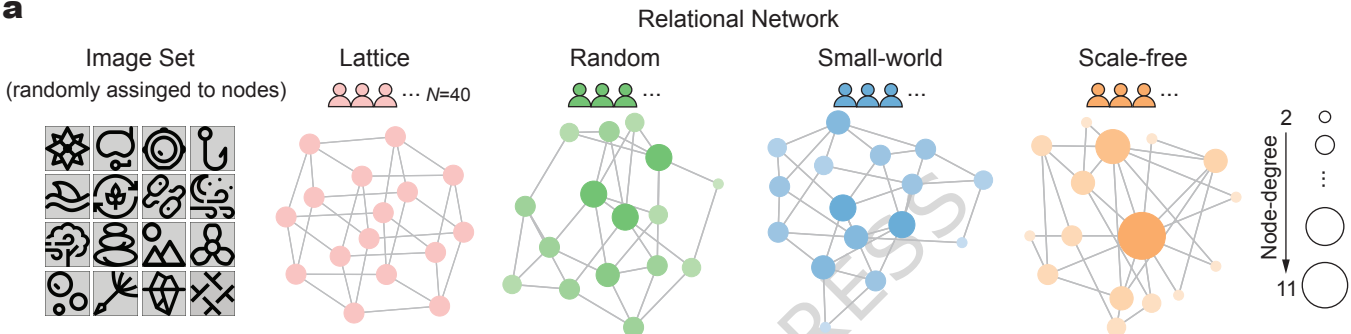
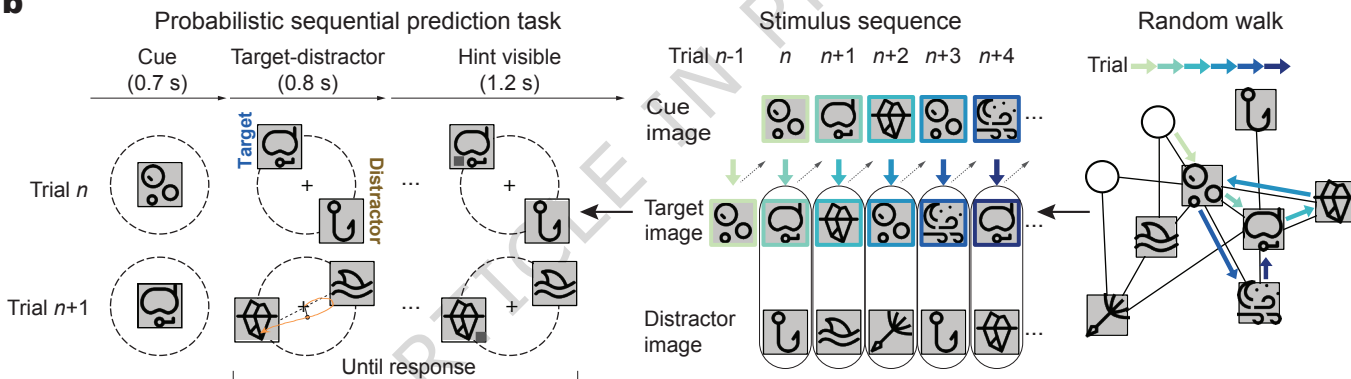
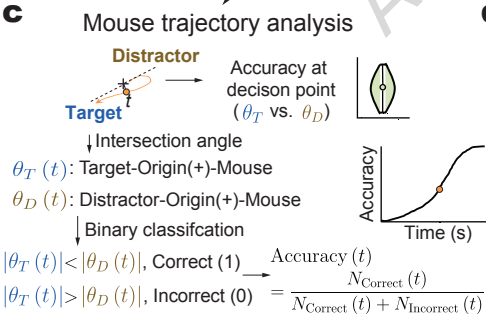
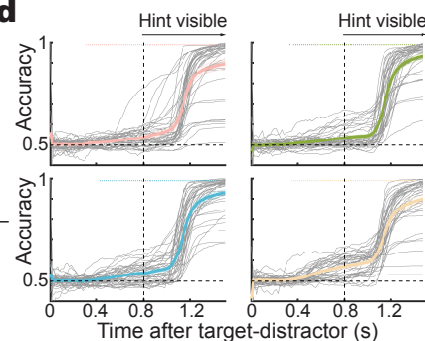


Network compressibility relates to inhomogeneity



Compressive learning based on inhomogeneity



a**b****c****d****e**